

Solution

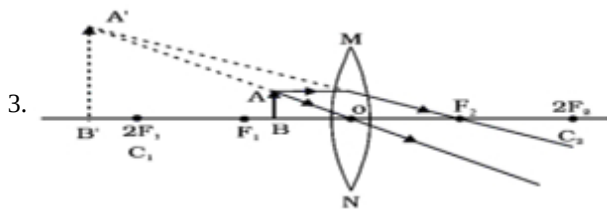
LIGHT -08

Class 10 - Science

Section A

1. Convergent

2. $f(m) = \frac{1}{P} = \frac{1}{2.5} = 0.4 \text{ m or } 40 \text{ cm}$



4. i. Convex lens - magnified

Concave lens - diminished

ii. Convex - object between O and F

Concave - object anywhere between optical centre and infinity.

5. According to Snell's law of refraction,

$$\frac{\sin i}{\sin r} > 1 \text{ or } \sin i > \sin r$$

$$i > r$$

hence, angle of incidence is greater than angle of reflection.

6. As, ${}_1\mu^2 = \frac{\sin i}{\sin r}$

$$\frac{\sin 45^\circ}{\sin 30^\circ} = \frac{1/\sqrt{2}}{1/2} = 1.41$$

7. As, $\mu_{\text{glass}} = 1.5$, $c = 3 \times 10^8 \text{ m/s}$

$$\therefore \mu = \frac{c}{v} \text{ or } 1.5 = \frac{3 \times 10^8}{v}$$

$$v = 2 \times 10^8 \text{ m/s}$$

8. Given, refractive index of a with respect to b is ${}_b\mu_a = 2$

$\therefore$ Refractive index of b with respect to a is

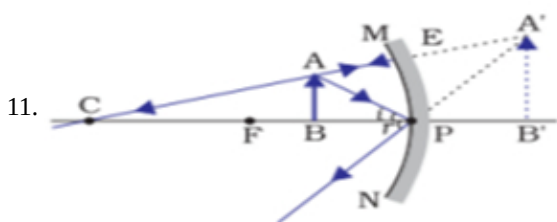
$$\frac{1}{{}_b\mu_a} = {}_a\mu_b = \frac{1}{2} = 0.5$$

9. Torches, search light, vehicles head lights, shaving mirrors, dentist's mirror, Solar furnaces.

10. $f = 15 \text{ cm}$

$$R = 2f$$

$$R = 2 \times 15 \text{ cm} = 30 \text{ cm}$$



12. i. $h = +10 \text{ cm}$

$$u = -100 \text{ cm}$$

$$v = -100 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{100} - \frac{1}{100} = \frac{1}{f}$$

$$\frac{-2}{100} = \frac{1}{f}$$

$$f = -50 \text{ cm}$$

ii. $m = \frac{-v}{u} = \frac{-(-100)}{100} = -1$

13. Given, $f = 20 \text{ cm}$, $u = -10 \text{ cm}$

$$\text{Using, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} - \left(-\frac{1}{10}\right) \Rightarrow v = -20 \text{ cm}$$

14. As, $m = \frac{v}{u} = \left(\frac{-20}{-10}\right) = 2$

$$\therefore m = \frac{h_2}{h_1}$$

$$2 = \frac{h_2}{4} \Rightarrow h_2 = 8 \text{ cm}$$

15. Here, $m = -2$

$$h_2 = -6 \text{ cm}$$

$$h_1 = ?$$

$$\text{As, } m = \frac{h_2}{h_1} \Rightarrow -2 = \frac{-6}{h_1} \Rightarrow h_1 = 3 \text{ cm}$$

16. As $P = \frac{1}{f}$ ($\because f = 5 \text{ cm}$)

$$P = \frac{-1}{0.05 \text{ m}} = -20 \text{ D}$$

Section B

17. State True or False:

(i) **(a)** True

Explanation: True

(ii) **(b)** False

Explanation: False. Convex mirrors are fitted on the sides of the vehicle, enabling the driver to see traffic behind him/her to facilitate safe driving.

(iii) **(a)** True

Explanation: True

(iv) **(a)** True

Explanation: True

(v) **(a)** True

Explanation: True

(vi) **(a)** True

Explanation: True

(vii) **(a)** True

Explanation: True

(viii) **(a)** True

Explanation: True

(ix) **(b)** False

Explanation: False. The SI unit of power of a lens is 'diopetre'.

(x) **(a)** True

Explanation: True

(xi) **(a)** True

Explanation: True

(xii) **(a)** True

Explanation: True

(xiii) **(a)** True

Explanation: True

(xiv) **(a)** True

Explanation: True

(xv) **(a)** True

Explanation: True

(xvi) **(b)** False

Explanation: False. Reflection is the phenomenon of bouncing back of light into the same medium on striking the surface of any object.

- (xvii) **(a)** True
Explanation: True
- (xviii) **(a)** True
Explanation: True
- (xix) **(a)** True
Explanation: True
- (xx) **(a)** True
Explanation: True

Section C

18. Fill in the blanks:

- (i) 1. Refraction
(ii) 1. Diverging
(iii) 1. Principal axis
(iv) 1. Refractive index
(v) 1. Power
(vi) 1. Normal
(vii) 1. Left
(viii) 1. Magnification
(ix) 1. Optical centre
(x) 1. Less
(xi) 1. Reflection
(xii) 1. Speed
(xiii) 1. Concave
(xiv) 1. Towards
(xv) 1. Negative
(xvi) 1. Virtual
(xvii) 1. Snell's
(xviii) 1. Dioptre

Section D

19.

(b) 1-A, 2-C, 3-B, 4-D

Explanation: Refraction is the phenomenon of change in the direction of propagation of light when it passes from one transparent medium into another. Reflection is the phenomenon of bouncing of light from the interface of an opaque medium. When a ray of light passes from a denser medium to a rarer medium, the velocity of light increases. When a ray of light passes from a rarer medium to a denser medium, the velocity of light decreases.

(1) Refraction	(A) Bending of light
(2) Reflection	(C) Bouncing back of light
(3) Rarer medium	(B) Velocity of light increases
(4) Denser medium	(D) Velocity of light decreases

20. **(a)** 1-B, 2-D, 3-A, 4-C

Explanation:

The virtual image formed by a plane mirror is virtual, erect and of the same size as that of the object. According to New Cartesian Sign Convention, the focal length (and power) of a convex lens is positive, and that of a concave lens is negative. A concave lens always produces erect and virtual images. A convex mirror always forms virtual images that are erect and diminished. The focal length of a convex mirror is positive, and that of a concave mirror is negative. A convex lens produces real and enlarged images when the object is placed beyond the centre of curvature.

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(1) Plane mirror	(B) Virtual image, erect and same size
(2) Concave lens	(D) Virtual image, negative focal length, negative power
(3) Convex lens	(A) Real image, inverted and large
(4) Convex mirror	(C) Virtual image, positive focal length

21.

(c) 1-B, 2-D, 3-A, 4-C

Explanation:

A normal is a perpendicular drawn to the reflecting surface at the point of incidence. Marginal rays are the rays that strike the surface of the mirror near its periphery. Linear magnification (m) is the ratio of the height of the image to the height of the object. It has no unit. The absolute refractive index of a medium is given by: $n_m = \frac{\text{speed of light in air}}{\text{speed of light in the medium}} = \frac{c}{v}$ where c is the speed of light in air, and v is the speed of light in the medium.

(1) A perpendicular drawn at the point of incidence to the reflecting surface	(B) Normal
(2) The rays which strike the mirror surface near the periphery	(D) Marginal ray
(3) The ratio of the height of the image to the height of the object	(A) Linear magnification
(4) The ratio of speed of light in vacuum to the speed of light in a medium	(C) Absolute refractive index

22. (a) 1-A, 2-C, 3-B, 4-D

Explanation:

The reflecting surface of a spherical mirror forms a part of a hollow sphere. The reflecting surface has a circular outline. The radius of the sphere of which the reflecting surface of the mirror forms a part is called the **radius of curvature** of the mirror. The **pole** is the midpoint of the reflecting surface of a spherical mirror. The diameter of the reflecting surface of a spherical mirror is called its **aperture**.

(1) Pole	(A) The midpoint of the reflecting surface of spherical mirror.
(2) Principal axis	(C) The imaginary straight line joining the pole and the centre of curvature.
(3) Radius of curvature	(B) The radius of hollow sphere of which the reflecting surface forms a part.
(4) Aperture	(D) The width of the mirror from which reflection can take place.

23. (a) 1-C, 2-B, 3-D, 4-A

Explanation:

The mirror formula gives the relationship between the focal length (f), the object distance (u) and the image distance (v) for a spherical mirror. The reciprocal of the focal length (f) is equal to the sum of the reciprocals of the object distance (u) and the image distance (v). For a spherical mirror of small aperture, the focal length (f) is one-half of the radius of curvature (R). The lens formula gives the relationship between the focal length (f), the object-distance (u) and the image-distance (v) for a lens. Magnification produced by a mirror/lens gives the relative extent to which the image of an object is magnified with respect to the object. It is the ratio of the height of the image to the height of the object. It is also related to the object-distance (u) and the image-distance (v).

(1) Mirror formula	(C) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
(2) Lens formula	(B) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
(3) Magnification	(D) $-\frac{v}{u}$
(4) Focal length	(A) $\frac{R}{2}$

24.

(b) 1-D, 2-A, 3-C, 4-B

Explanation:

A convex lens is a converging lens; a concave lens is a diverging lens. A convex mirror always forms a virtual and a diminished image, for any position of the object. The virtual image formed by a concave mirror (when the object is between the pole and the focus) is larger than the object.

(1) Convex lens	(D) Converging
(2) Concave lens	(A) Diverging
(3) Concave mirror	(C) Virtual image larger than the object
(4) Convex mirror	(B) Virtual image smaller than the object

25.

(c) 1-D, 2-A, 3-C, 4-B

Explanation:

A **concave mirror** is a spherical mirror whose reflecting surface is curved inwards. A **convex mirror** is a spherical mirror whose reflecting surface is curved outwards. **Lateral inversion** is the property of plane mirrors where objects actually to the left appear to be on the right in the image. **Focal length** is the distance between the pole (P) and the focal point (F) of a mirror/lens.

26. (a) 1-C, 2-B, 3-D, 4-A

Explanation: Diamond has a high refractive index of 2.42; ice has refractive index of 1.31. In the head lights of vehicles, the reflectors are concave in shape and the bulb is placed very near to the focus; the reflected light forms a powerful parallel beam and travels a large distance. Convex mirrors always form erect and diminished images and give a wide field of view; they are used as rear-view mirrors or wing mirrors. These enable the driver of the vehicle to see the traffic behind the vehicle to facilitate safe driving.

(1) Highest refractive index	(C) Diamond
(2) Lowest refractive index	(B) Ice
(3) Head-lights of a car	(D) Concave mirror
(4) Side rear view of a mirror	(A) Convex mirror

Section E

27.

(c) lens and screen only

Explanation: Parallel rays get converged at the focus.

28.

(c) Larger

Explanation: When a lemon is kept in water in a bowl and is viewed from the outside, it appears larger than its actual size due to refraction. When reflected rays of light travel from the lemon (which is in water) towards the eye (which is in the air), the rays travel from a denser medium to a rarer medium and they bend away from the normal to the water surface.

29.

(d) less than 1

Explanation: The convex mirror forms virtual, erect, and diminished image of the object and rearview mirror also form some type of image. Therefore, magnification (m) produced by a rearview mirror fitted in vehicles is less than one.

Section F

30. (a) C, D, B, A

Explanation: Unless an object is chosen and the setting of lens and screen is proper, image distance and thereby the focal length cannot be found.

Section G

31. (a) Both statements are correct.

Explanation: Both statements are correct. A real image is always inverted. A virtual image is always erect.

Section H

32. i. (c) less than 1, more than 1 or equal to 1

ii. (d) equal to one

iii. (b) at the centre of curvature

iv. (a) magnification

- v. (a) virtual; erect
- 33.
 - i. (a) parallel to incidence ray
 - ii. (b) refracted away from the normal
 - iii. (b) from rarer to denser medium
 - iv. (a) air
 - v. (b) greater
- 34.
 - a. (d) S
 - b. (a) absolute refractive index
 - c. (a) 1.33
 - d. (a) 3×10^8 m/s
 - e. (b) more
- 35.
 - i. (d) At the focus F
 - ii. (b) At C
 - iii. (c) Both (a) and (b)
 - iv. (a) at the focus F
 - v. (c) Highly enlarged
- 36.
 - i. (c) spectacles for the correction of short sight
 - ii. (b) virtual, erect and diminished
 - iii. (a) concave lens
 - iv. (b) focal length
 - v. (a) converges, diverges